Physical Leaves Simulator — User Guide

By Vladislav (Lazy3D)

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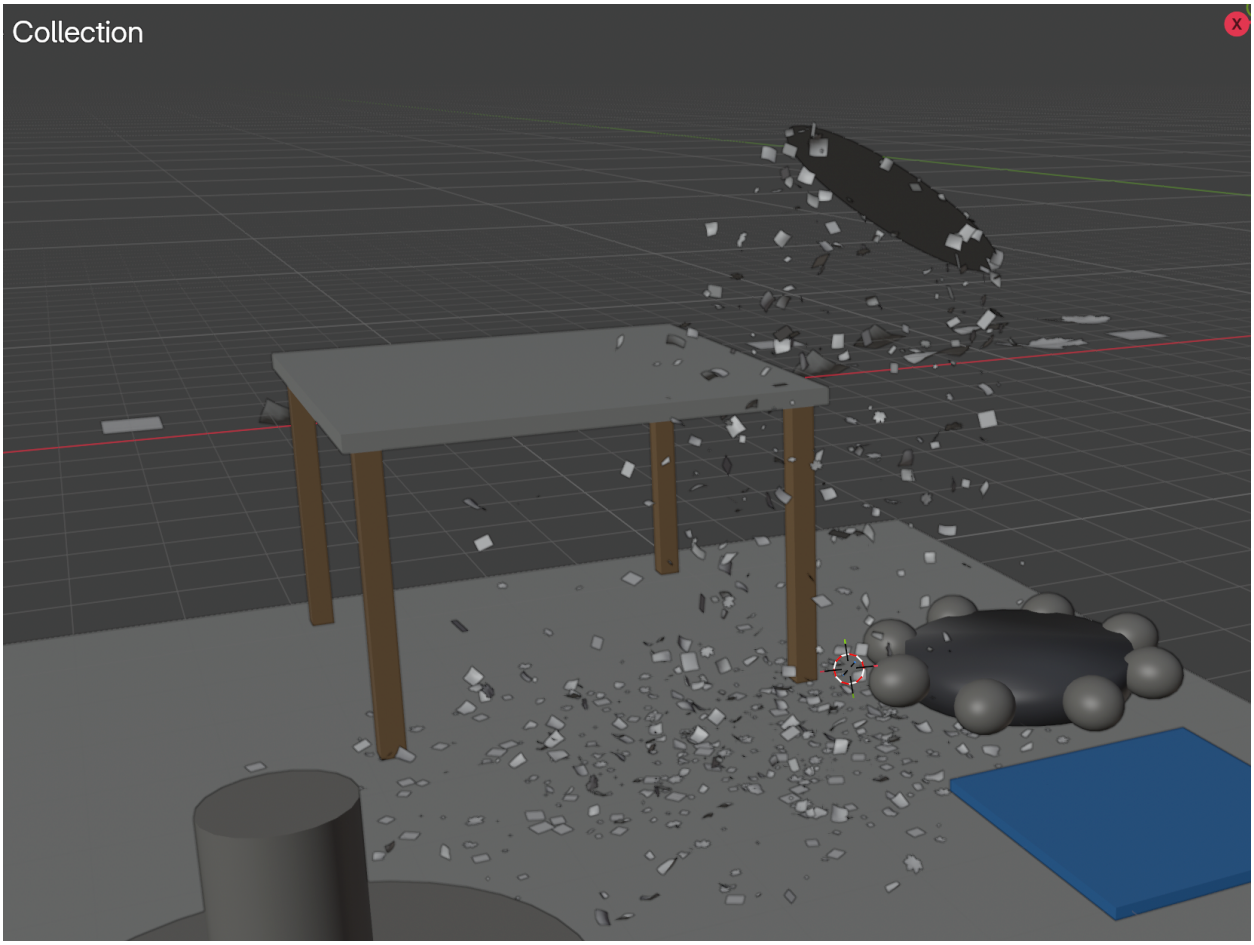
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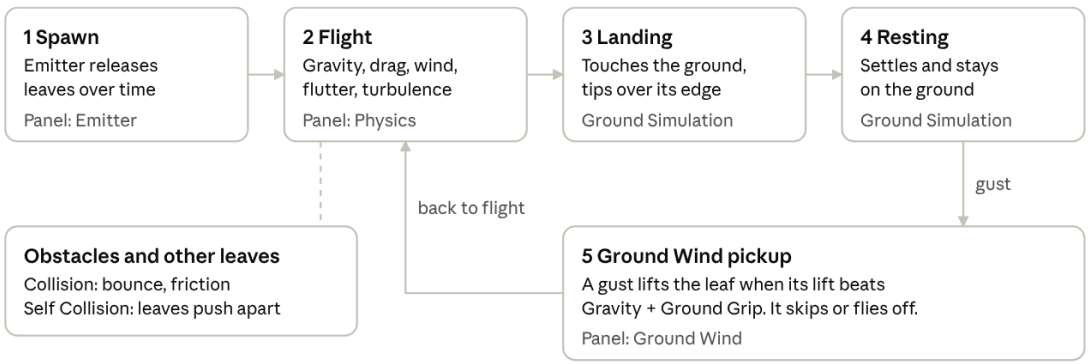
How it works

Physical Leaves drops leaves from an emitter, flies them through wind, lands them on your ground and lets gusts lift them again. Every leaf goes through the same stages, and each stage has its own panel in **Simulator Settings**.

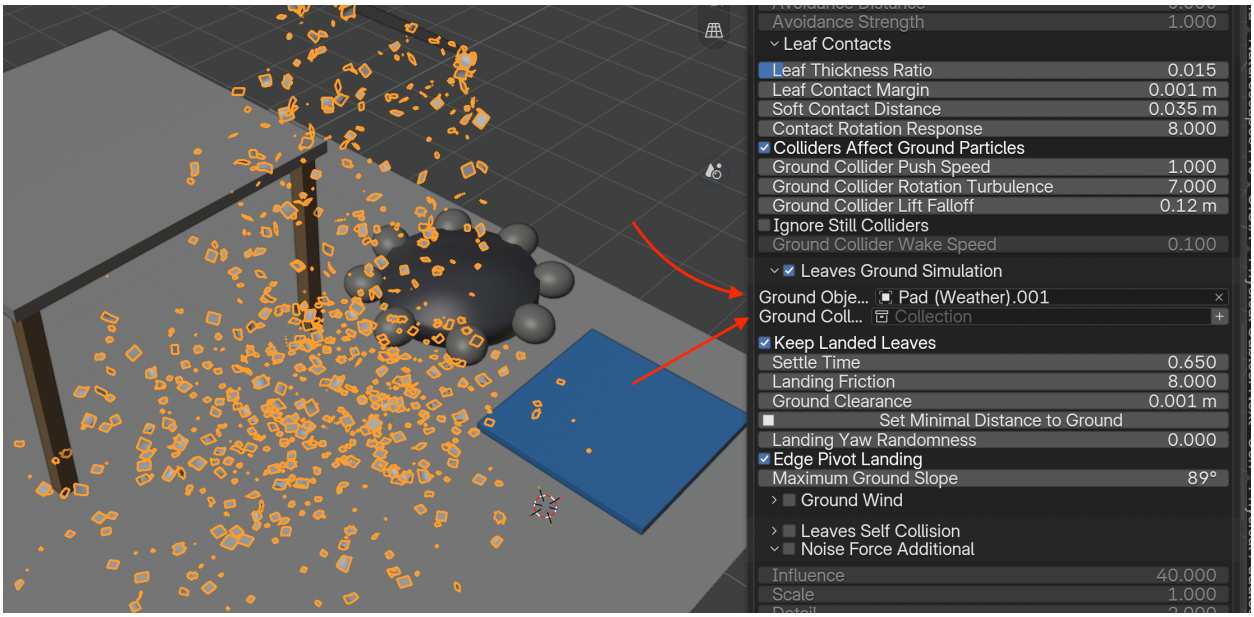


A leaf falls, lands, rests — and a gust can lift it again

Each stage of a leaf's life and the panel that controls it

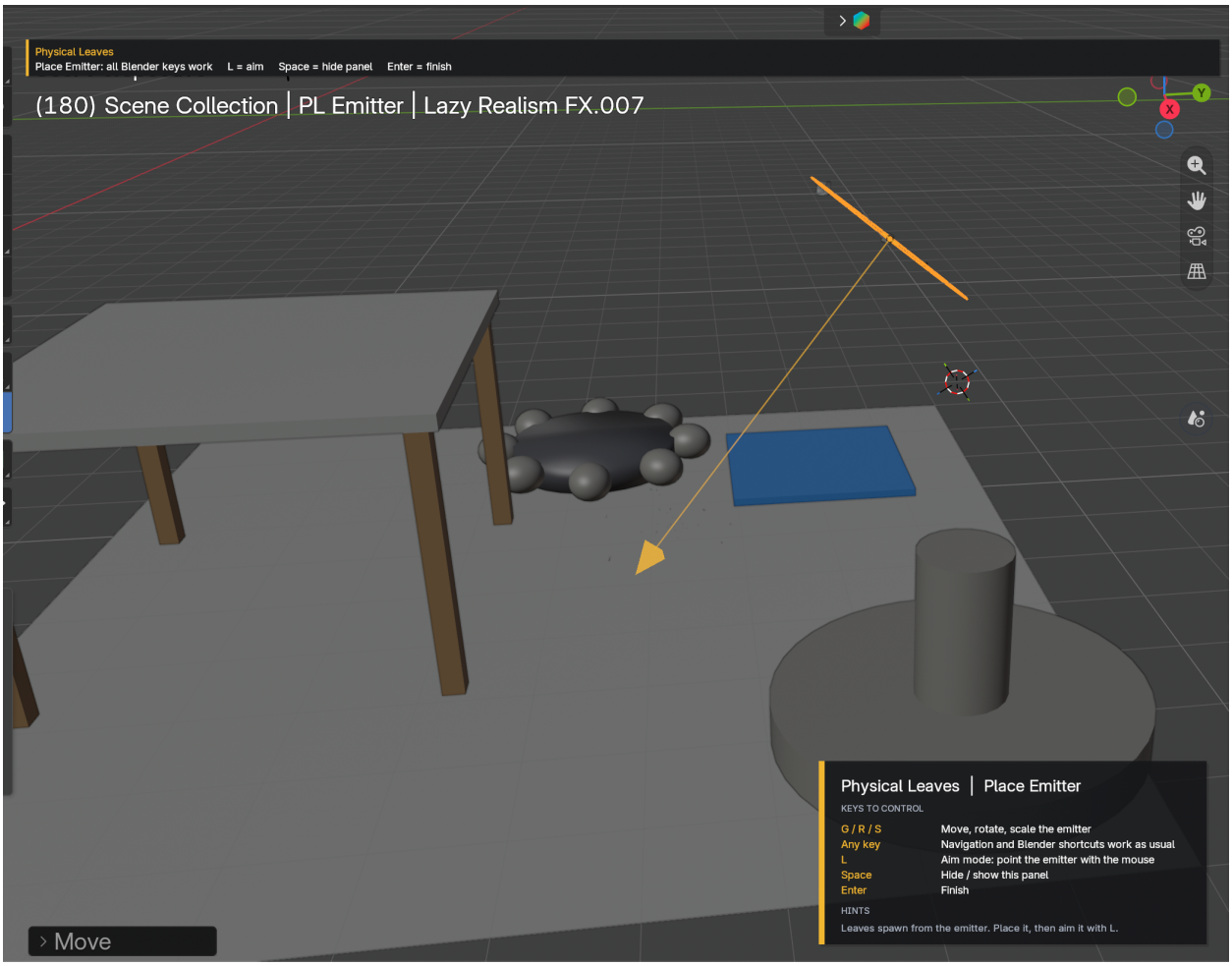


A leaf only lands on the Ground Object or Ground Collection you set. Obstacles and other leaves affect it in flight and on the ground.

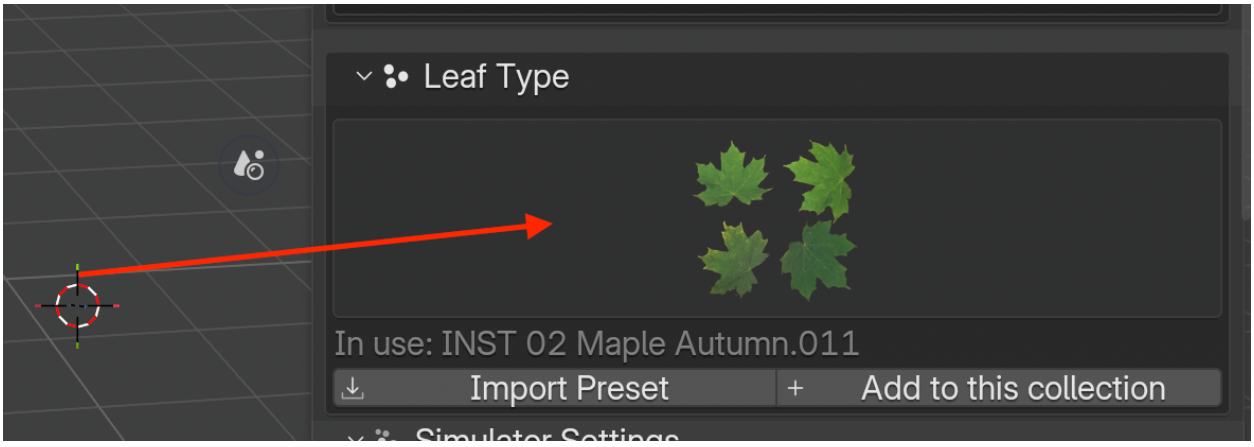


Quick start

1. Click **Physical Leaves**. The simulator and its emitter appear at the 3D Cursor.
2. Move the emitter with **G / R / S**. Press **L** and point the mouse where the leaves should fly: the orange arrow shows their path. Click to keep the direction, **Enter** to finish.



3. In **Simulator Settings**, set **Ground Object** or **Ground Collection** — the surface where leaves land.
4. In **Leaf Type**, pick a leaf and click **Import Preset** (a maple leaf is assigned by default). To use your own leaves, see *Your own leaves*.

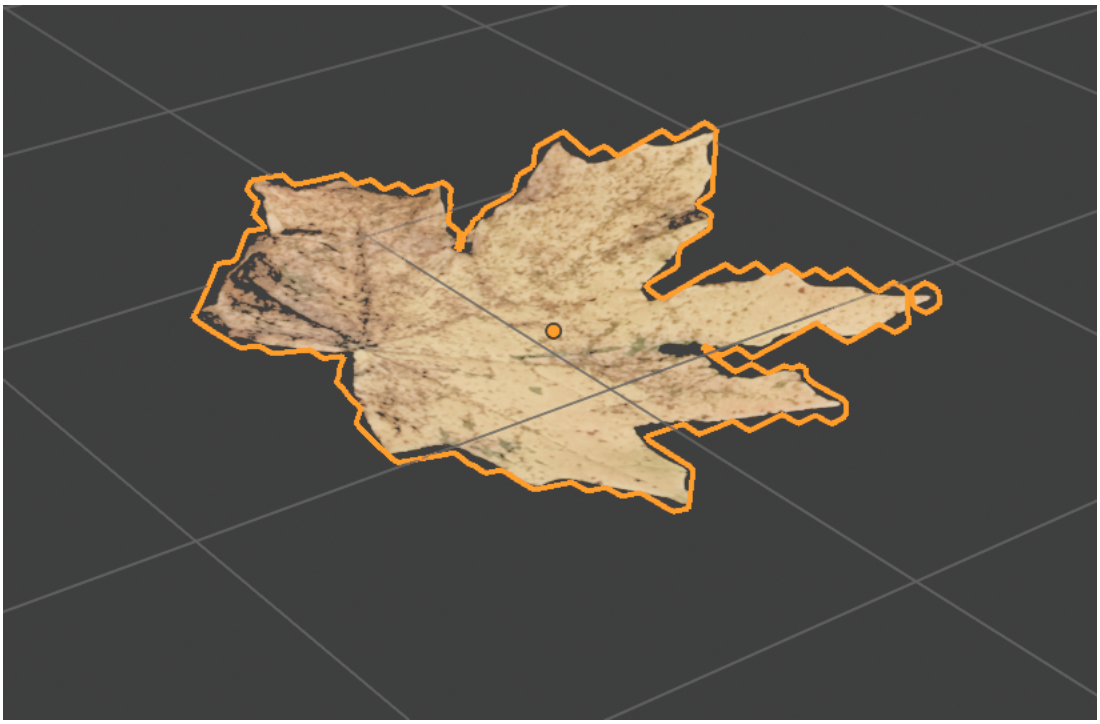


5. Open **Presets** and click one to start from a tuned setup, for example *03 Gentle Breeze*.
6. Press Play to preview. Then save the .blend and click **Bake Leaves**.

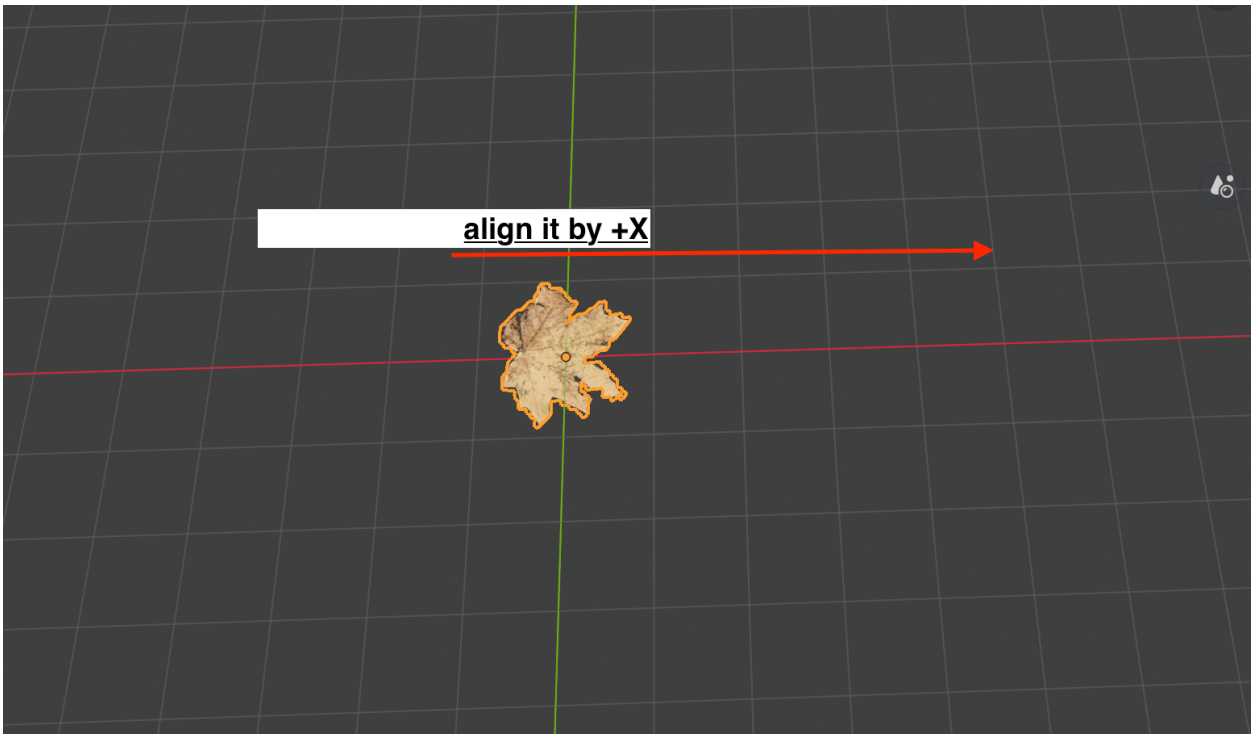
Your own leaves

For every contact the simulator replaces each leaf with a flat disk, measured from your leaf mesh. Prepare each leaf object so the disk matches the leaf, then put it in **Instance Object** or **Instance Collection** (panel Instances).

1. **Lay the leaf flat in XY, face up (+Z).** The disk radius comes from X and Y, its thickness from Z.



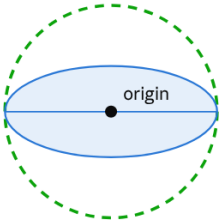
2. **Put the origin in the leaf's centre:** Object → Set Origin → Origin to Geometry.
3. **Apply rotation and scale: Ctrl+A → Rotation & Scale.** The simulator uses the mesh only and ignores object transforms.
4. **Fix orientation in the mesh, not with Offset Rotation.** Offset Rotation turns only the visible leaf, not its disk.



Centre the origin so the disk matches the leaf

Top view. Dashed circle = the contact disk the simulator uses.

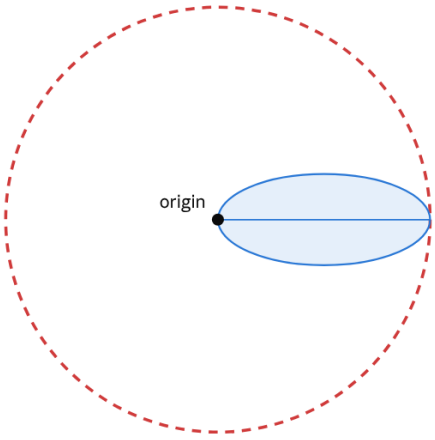
Correct: origin in the centre



Radius = half the leaf length

Disk matches the leaf

Wrong: origin at the stem



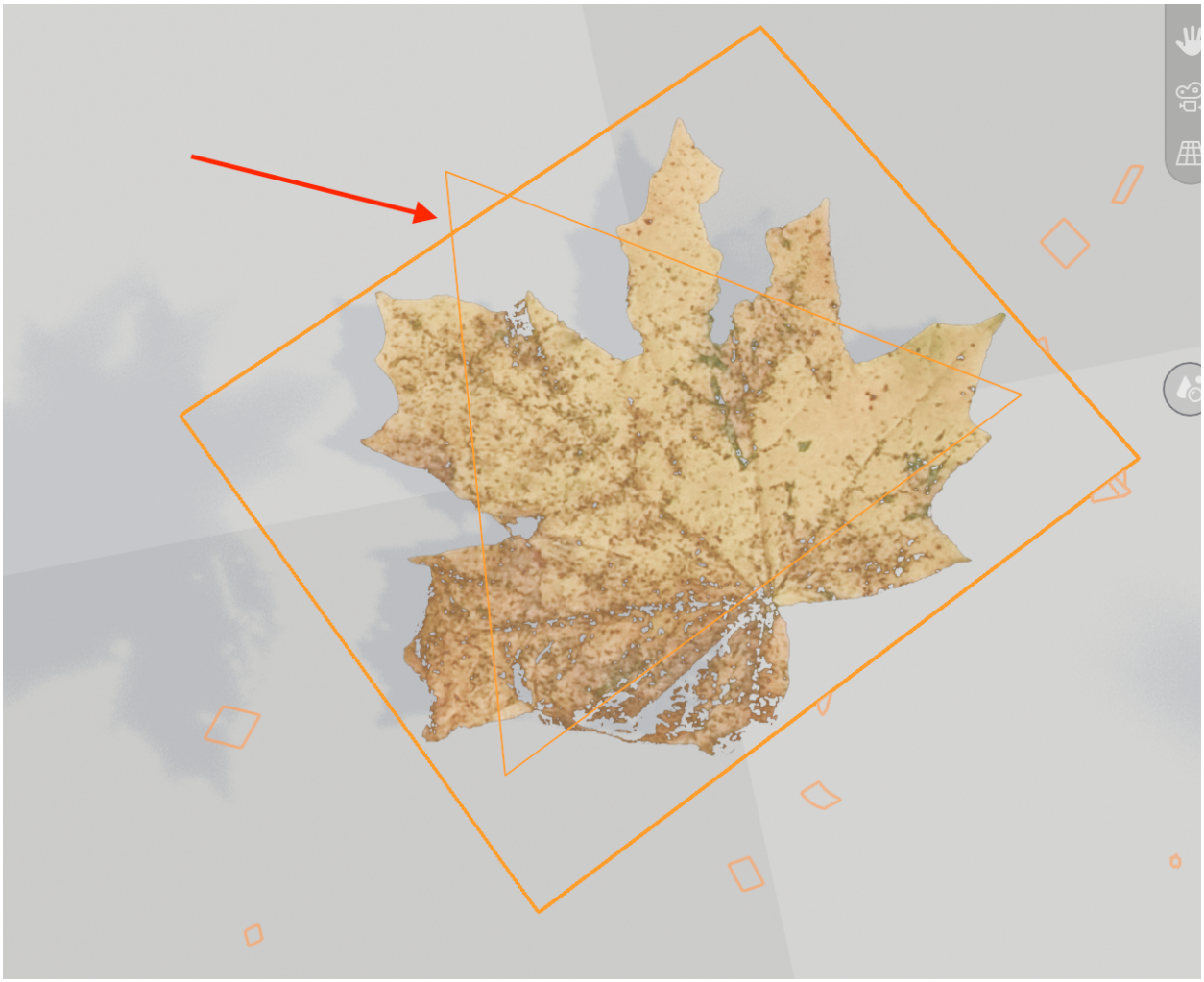
Radius = full leaf length (2× too big)

Disk is offset: hits ground and neighbours early

The disk is measured from the origin to the farthest corner of the leaf, so an origin at the stem makes it twice too big and shifts it off the leaf.

Leaf size inside the simulator = mesh size × Instance Scale × particle size (Master Size with Randomize Size, used when Size Scale is on). The disk follows the same size; Physical Bounds Radius scales the disk alone.

Size does not change how leaves fly. It changes how they rest on slopes and edges and how far apart they pile up. Turn on **Show Physical Bounds** to check: each disk outline should cover its leaf.



Physics and flight

A falling leaf is pulled down by Gravity and held back by Air Drag. Its steady falling speed is:

Falling speed = √(|Gravity| ÷ Air Drag)

Lower Gravity slows the whole motion; real gravity is −9.81. None of these controls depend on leaf size: a big leaf flies exactly like a small one.

Simulation

Control	What it does
Time Scale	Speed of simulation time. Below 1 = slow motion; 1 = normal; above 1 = faster. 0 pauses the motion.
Substep Iterations	Physics steps per frame. More steps make fast motion and

Control	What it does
	collisions more stable but cost more time.

Velocity and wind

Control	What it does
Gravity	Pulls leaves down (negative Z). A larger value falls faster and feels heavier; a smaller one feels lighter. X and Y add a sideways pull.
Normal Velocity	Launch speed along the emitter's surface normal. Positive values push leaves out; negative values launch them the opposite way.
Random Velocity	Adds a random launch speed in all directions. Higher values give a wider, faster spread.
Wind (checkbox)	Turns the leaves' response to the base wind on or off. Off = drag, lift and flutter ignore the wind.
Wind	Base wind velocity: the vector sets the direction, its length the strength. Around 1 is gentle to moderate.
Use Wind Direction Controls	Replaces the Wind vector with Wind Direction, Wind Elevation and Wind Strength.
Wind Direction	Horizontal direction: 0° = +X, 90° = +Y.
Wind Elevation	Vertical tilt: 0° = horizontal, positive = upward.
Wind Strength	Wind speed; 0 = no wind.
Show Wind Direction	Draws the wind as an arrow in the viewport; Wind Arrow Length and Offset set its size and place.

Aerodynamics Advanced

Control	What it does
Air Drag	Air resistance when the leaf falls flat. Higher values brake harder and slow the fall; lower values let leaves fall and drift freely.
Edge Drag	Air resistance when the leaf moves edge-first. Lower values let edge-on leaves slice through the air; higher values slow them.
Drag Randomness	Gives each leaf its own drag. 0 = all leaves alike; higher values make falls and drifts more varied.
Leaf Lift	Lift from the airflow. 0 = none; higher values hold the leaf up so

Control	What it does
	it glides and floats longer.
Flutter Strength	Strength of the rocking motion. Low = subtle rocking; high = strong flips and swings.
Flutter Frequency	How fast the flutter changes. Low = slow, broad rocking; high = quick fluttering.
Spin Speed	How much leaves spin in the air. Higher = faster continuous spin.
Air Alignment	Turns the leaf toward its balanced position in the airflow. Higher = turns faster and harder; 0 = tumbles freely.
Angular Damping	Slows rotation down. Low = spins and tumbles last longer; high = rotation settles sooner.
Maximum Leaf Speed	Hard speed limit. Lower values keep all leaves slower.

Flight Turbulence

Control	What it does
Flight Turbulence (checkbox)	Adds random motion to flying leaves. Off = only aerodynamics and forces move them.
Flight Turbulence Strength	How far turbulence pushes leaves off their path.
Flight Turbulence Frequency	How quickly turbulence changes. Low = broad, slow wandering; high = fast, busy motion.
Flight Rotation Turbulence	Random rotation in the air. Higher = stronger, irregular spinning and tumbling.
Flight Turbulence Fade In	Time for turbulence to build up after a leaf takes off. 0 = full turbulence at once.

Ground and collisions

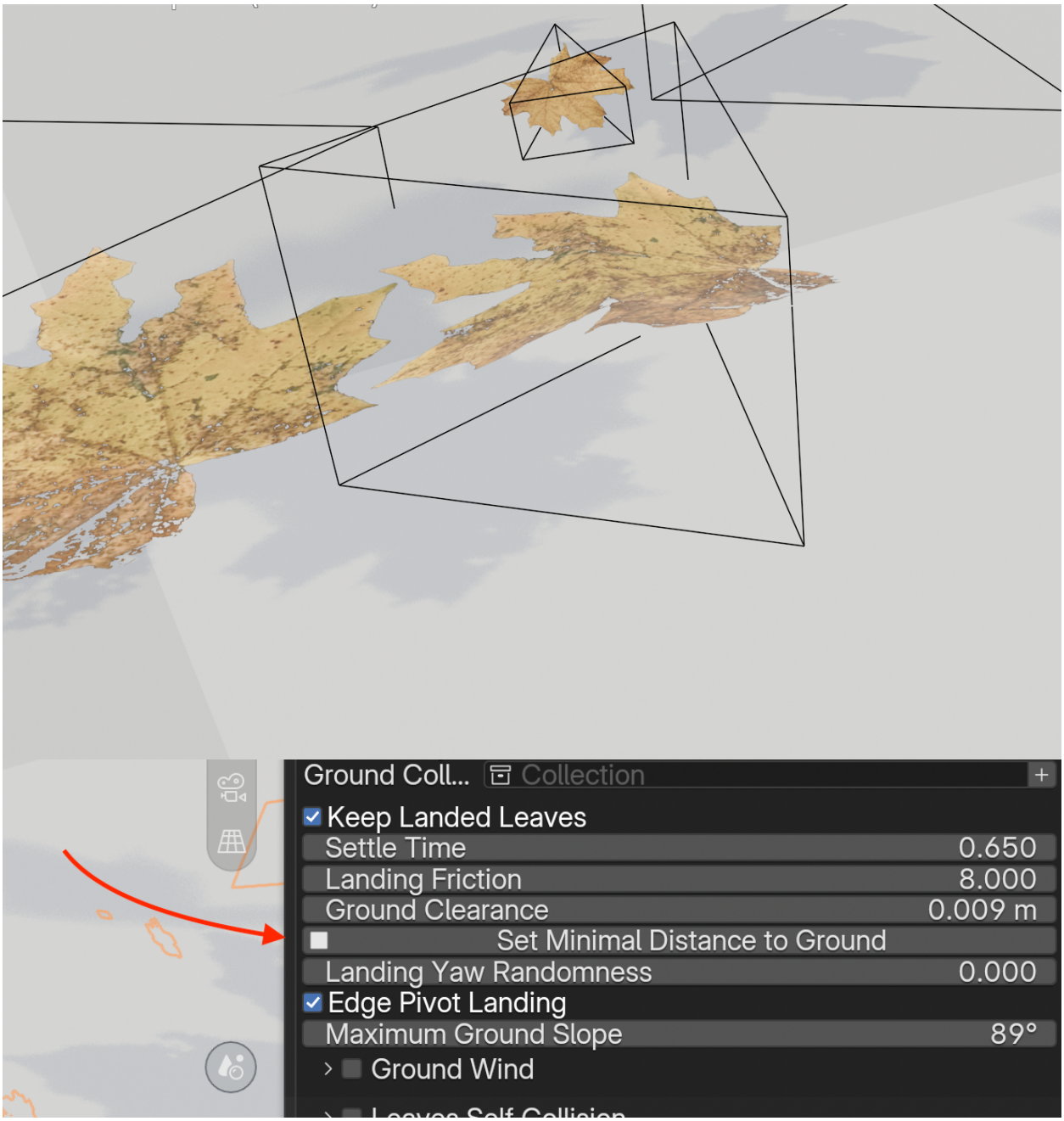
Leaves land on the Ground Object or Ground Collection, settle, and stay until wind, colliders or other leaves wake them. Each module has a checkbox on its panel header; a module that is off skips its calculations.

Leaves Ground Simulation

Control	What it does
Enable Ground Simulation (checkbox)	Turns landing on or off. Off = leaves never land, no ground geometry is loaded, and Ground Wind switches off too.

Control	What it does
Ground Object	The surface leaves land and rest on; it also gives the ground direction for settling. Empty = no ground from an object.
Ground Collection	A collection of ground surfaces, used with or instead of Ground Object.
Keep Landed Leaves	Keeps landed leaves on the ground for good, so wind, colliders or other leaves can wake them. Off = they still disappear when Particle Life ends.
Settle Time	Length of the landing transition. Lower = leaves come to rest sooner; higher = they slide and settle longer.
Landing Friction	Braking along the ground while landing. Lower = leaves slide further; higher = they stop sooner.
Ground Clearance	Small gap between a resting leaf and the ground. 0 = right on the surface; higher values avoid clipping but can look like floating.
Set Minimal Distance to Ground (button)	Sets Bounds Height to 0 and Ground Clearance to 0.001, so leaves rest as close to the ground as possible.
Landing Yaw Randomness	Random turn around the ground normal when a leaf settles. Higher = more varied leaf directions on the ground.
Edge Pivot Landing	On: the edge that touches first stays planted and the leaf tips over it. Off: the leaf turns around its centre (older behaviour).
Maximum Ground Slope	Steepest surface that still counts as ground. Low angles accept only flat ground; near 90° almost vertical faces count.

Leaves hover above the ground? Click **Set Minimal Distance to Ground** in Leaves Ground Simulation. It sets Bounds Height to 0 and Ground Clearance to 0.001.


Ground Coll...
Collection
Keep Landed Leaves
Settle Time
0.650
Landing Friction
8.000
Ground Clearance
0.009 m
Set Minimal Distance to Ground
Landing Yaw Randomness
0.000
Edge Pivot Landing
Maximum Ground Slope
89°
Ground Wind



Collision

Control	What it does
Enable Collision (checkbox)	Turns collisions with the Collider collection on or off. Ground landing is separate.
Collider	Collection of obstacle objects. More or heavier meshes cost more time; empty = no obstacles.
Bounce	How much speed bounces back after a hit. 0 = soft, dead impact; 1 = strongest rebound.
Friction	How much sliding speed a hit removes. 0 = slippery; 1 = strongest grip.
Delete on Hit	Removes a leaf as soon as it hits an obstacle.
Fall Off Edges	On: leaves slide off the border of the ground mesh and fall. Off: leaves near an open edge are held there (older behaviour).
Avoid Colliders	Flying leaves steer smoothly around the Collider collection instead of hitting it, and turn smoothly toward the surface instead of flipping on a hit; Contact Rotation Response sets how fast. Landed leaves are unchanged. Off = normal collisions.
Avoidance Distance	How far from a collider flying leaves start to steer away. Larger = earlier, wider curves.
Avoidance Strength	How firmly leaves turn away inside the Avoidance Distance. Low = late, gentle curves; high = early, firm turns.

Leaf Contacts

Control	What it does
Leaf Thickness Ratio	Thickness of the leaf's contact disk relative to its radius. Larger = leaves keep further apart along their normal.
Leaf Contact Margin	Extra safety distance for detecting contacts. Very small values allow close contact but risk clipping; larger values can make leaves hover apart.
Soft Contact Distance	Distance at which a contact starts softening before touching. Larger = smoother but floatier; smaller = later, harder contact.
Contact Rotation Response	How strongly a contact turns the leaf. Lower = keeps its orientation; higher = rotates more on impact. With Avoid Colliders it sets how fast avoiding leaves line up with the surface; 0 = no turning.
Colliders Affect Ground Particles	Lets moving colliders push and wake leaves that have already landed.
Ground Collider Push Speed	Push speed given to a landed leaf woken by a collider.
Ground Collider Rotation Turbulence	Random spin given to a landed leaf woken by a collider.
Ground Collider Lift Falloff	Distance over which a collider's lift fades around a landed leaf. Small = tight to the contact; large = wider, softer effect.
Ground Collider Wake Speed	How fast a collider must move toward a landed leaf to wake it. A collider that stands still never wakes leaves, so leaves next to it stay calm. 0 = any movement wakes.
Ignore Still Colliders	On (default): a collider that stands still never wakes landed leaves, so they don't flicker next to it; only one moving faster than Ground Collider Wake Speed wakes them. Off: every landed leaf near a collider is woken, also by a still one.

Leaves Self Collision

Control	What it does
Self Collision (checkbox)	Leaves collide with each other: flying leaves with flying and landed ones. Off = no leaf-to-leaf contact.
Even Ground	Also separates two leaves that both lie on the ground, for even piles.
SC Iterations	Self-collision passes per substep. More = less overlap, but slower.

Control	What it does
SC Radius Scale	Collision radius relative to the leaf radius. Smaller = leaves can overlap more; larger = they keep further apart. Visible size is unchanged.
SC Strength	Strength of the separation. 0 = off; 1 = full.
SC Wake Speed	Smallest impact that wakes a landed leaf. Lower = leaves wake easily.

Ground Wind

Ground Wind lifts landed leaves in gusts, carries them along the ground and lets some fly off again. It needs a base Wind above zero: gusts vary the Wind, they cannot create it. For every Min / Max pair, each leaf gets its own random value in that range.

When does a landed leaf take off? When the wind's lift near the ground beats Gravity + GW Ground Grip. For more pickup, raise Wind or GW Lift Coefficient, or lower GW Ground Grip.

Main controls

Control	What it does
Enable Ground Wind (checkbox)	Turns ground pickup and gusts on or off.
Ground Object / Ground Collection	The same ground as in Leaves Ground Simulation, repeated here for quick access.
Affected Leaves	Share of leaves the wind may pick up, chosen once per leaf. 0 = none; 0.5 = about half; 1 = all. It does not change wind strength.
Wind Ground Object	Fallback ground for Ground Wind when no Ground Object or Collection is set.

Gust Field

Control	What it does
GW Gust Strength	How much gusts vary the wind speed. 0 = steady wind; around 1 = strong variation.
GW Gust Size	Size of gust patches. Small = local patches where neighbours differ; large = broad gusts that move many leaves together.
GW Gust Duration	How long gusts last, in seconds. Low = gusts change quickly; high = they persist and evolve slowly.
GW Direction Variation	How far gusts wander sideways from the wind direction. 0 =

Control	What it does
	straight; 1 = widest wandering.
GW Seed	Gives a different gust pattern; it is not stronger or weaker.

Surface Pickup

Control	What it does
GW Lift Coefficient	Lift the wind produces on a landed leaf. Higher = leaves lift more easily and faster; 0 = no pickup lift.
GW Ground Grip	Extra hold a landed leaf must overcome before it lifts. Higher = leaves cling to the ground.
GW Leaf Variation	Differences between leaves in response, grip and recovery. 0 = all leaves alike.
GW Response Time	How smoothly leaves react to wind changes. Low = almost instant; high = slower, smoother pickup and release.
GW Rest Time	Minimum rest on the ground before a leaf can be picked up again.

Lift and Carry

Control	What it does
GW Lift Duration Min / Max	How long the pickup push lasts. Short = quick kicks; long = sustained lifts.
GW Lift Height Min / Max	Height over which the pickup lift fades. Higher = leaves rise further. It is not a hard ceiling.
GW Ground Carry Min / Max	How strongly leaves near the ground follow the wind sideways. Higher = they travel further.
GW Tumble Min / Max	Rotation kick at pickup. Higher = stronger tumbling.

Continued Flight

Control	What it does
Flying Leaves Probability	Share of picked-up leaves that keep flying instead of settling again. 0 = none; 1 = all.
GW Flying Height Min / Max	Target height above the ground for flying leaves.
GW Flying Lift Min / Max	How firmly flying leaves hold their target height. Low = loose;

Control	What it does
	high = they return to it quickly.

Wind Rotation

Control	What it does
GW Rotation Strength	How strongly lifted leaves turn toward the airflow. 0 = no turning.
GW Rotation Speed	Fastest turn toward the airflow, in radians per second.

Wind Fade

Control	What it does
Wind Fade (checkbox)	Each pickup fades out after a while so gravity can settle the leaf again. Off = Ground Wind can keep a leaf flying forever.
GW Fade Delay Min / Max	Full-strength time after pickup before the fade starts, in seconds.
GW Fade Duration Min / Max	Time the fade takes to reach zero. Longer = gentler release.
GW Fade Falloff	Shape of the fade. 1 = default; below 1 keeps the wind longer; above 1 drops it earlier.

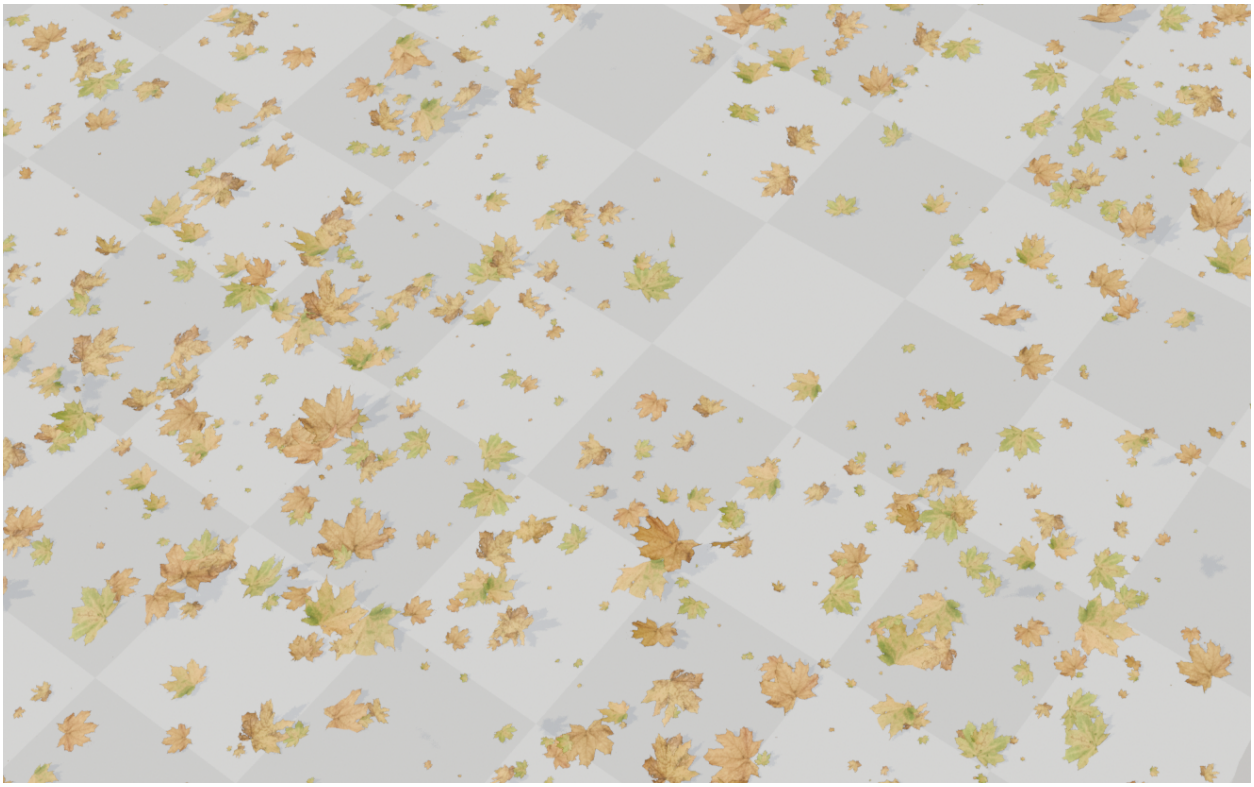
Most Common Problems

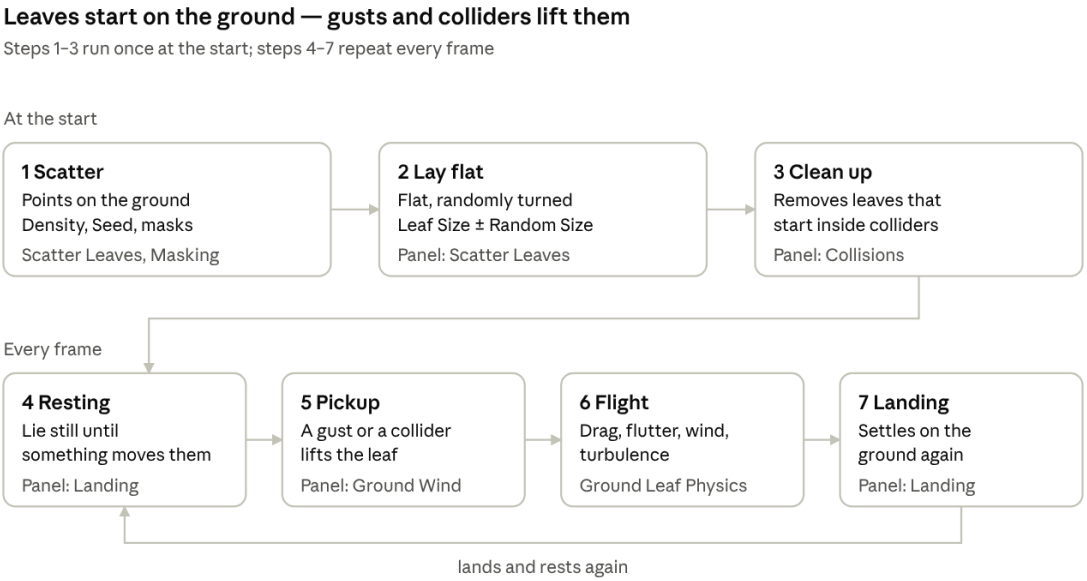
Problem	Fix
Leaves fall through the ground	Set Ground Object or Ground Collection; keep Leaves Ground Simulation on; check Maximum Ground Slope
Leaves hover above the ground	Click Set Minimal Distance to Ground; check the leaf origin is centred and the leaf lies in XY
Leaves are too big or too small	Change Instance Scale in the modifier
Landed leaves never lift	Raise Wind or GW Lift Coefficient; lower GW Ground Grip; raise Affected Leaves
Simulation is slow	Turn off Self Collision; lower Particles per Second or Max Leaf Count; use 2–3 Substep Iterations for previews
Changes do nothing	Delete the bake (trash icon next to Bake Leaves), go to frame 1, bake again

Problem	Fix
Bake Leaves is greyed out	Save the .blend file first
A moving object doesn’t push the landed leaves	Put the ground in the Ground Collection, not the Collider collection, and turn on Colliders Affect Ground Particles. See Ground vs Colliders.

Ground Leaves

Ground Leaves is the second leaf effect: instead of an emitter, it covers your ground with leaves that are already lying there. They rest until a wind gust or a moving collider (a character, a car, a ball) lifts them; then they fly with the same physics as Physical Leaves and land again.





The leaves never age out: they stay in the scene for the whole animation. Leaves land back on the same ground they were scattered on.

Ground Leaves setup

The add-on wires everything for you in one click. Everything is in **Sidebar (N) → Lazy Realism FX → Ground Leaves**.

1. Select the ground mesh so it is the **active** object. It goes straight into **Ground Object**: no collection is created and nothing is moved.
2. Click **Scatter** to cover the whole ground, or **Draw** to place leaves only where you paint.
3. With **Draw**, Weight Paint starts on the ground with a new vertex group. Paint where leaves should lie, then go back to Object Mode. Use **Paint Leaves (Weight Paint)** in the Masking panel to repaint later.
4. In **Leaf Type**, pick a leaf and click **Import Preset** (a maple leaf is assigned by default).
5. Open the **Wind** panel, tick its checkbox and set **Angle**, **Lift** and **Intensity**. Turn on **Wind Visualizer** to see the wind as an arrow above the ground.
6. Optional: to let objects push leaves (a character, a ball, a car), put them in a collection and assign it as **Collider** in the **Collisions** panel. Animate them through the leaves. Press Play to preview, then save the .blend and click **Bake Leaves**.

Several ground objects: turn on **Use Ground Collection** in Ground Scatter and assign a collection instead of the single Ground Object. See Ground vs Colliders below.

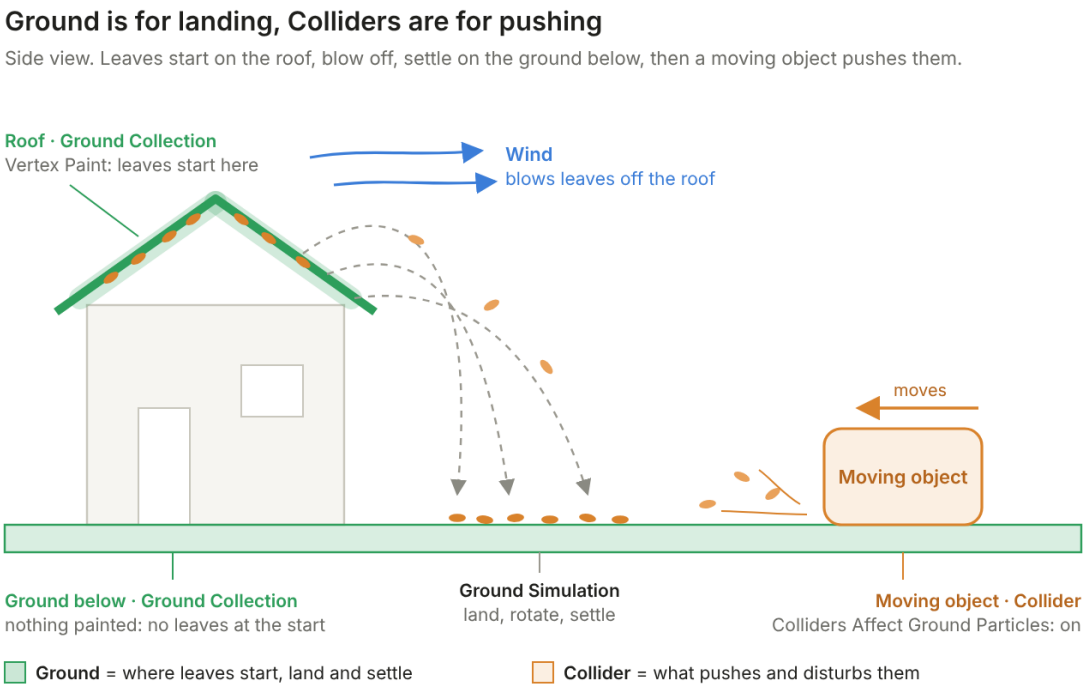
Getting the leaf size right. In Ground Leaves the contact disk is sized by **Leaf Size**, not by your leaf mesh. Turn on **Show Physical Bounds** and adjust **Physical Bounds Radius** until each circle covers its leaf. **Instance Scale → Apply Scale** resizes only the visible leaves, so check the circles again after using it. Own leaf meshes follow the same rules as in *Your own leaves*: flat in XY, origin in the centre, Ctrl+A applied.

Ground vs Colliders

One of the most important things to understand about Ground Leaves is the difference between Ground and Colliders: **Ground is for landing and settling, Colliders are mainly for pushing and disturbing leaves.** The same split applies in Physical Leaves: the Ground Object / Ground Collection is where leaves land, the Collider collection holds obstacles.

A practical example: leaves blown off a roof

Imagine you have some leaves scattered across a roof. You want the wind to blow them off the roof, let them fall naturally onto the ground below, and then let another moving object interact with those fallen leaves. To make this work, set up your Ground and Collider collections like this:



Setup

1. Assign **your roof and the ground below to the Ground Collection**, not the Collider collection. In Ground Scatter, turn on **Use Ground Collection** and pick that collection.
2. In Masking, click **Paint Leaves (Weight Paint)**, choose the roof and paint the leaf distribution on it. Don't paint anything on the ground below: no leaves should appear there at the start.
3. Put the moving object in the **Collider** collection (Collisions panel) and make sure **Colliders Affect Ground Particles** is enabled.
4. Play the simulation.

What happens

When you play the simulation, the leaves start on the roof and the wind gradually blows them away. Once they fall onto the ground, the dedicated Ground Simulation takes over: the leaves land, rotate and settle naturally on the surface.

Because the ground belongs to the Ground system, the separate moving collider can push away or disturb the leaves that have already landed. For example, an object moving through the fallen leaves pushes them aside as it passes.

What to avoid

- **Don't use the ground as a collider.** If the ground is in the Collider collection, the landed leaves won't respond to another collider in the same way.

Ground Leaves settings

The panels below are unique to Ground Leaves. The flight, landing, collision and Ground Wind panels share their controls with Physical Leaves (see the last table).

General

Control	What it does
Max Leaf Count	Upper limit on the number of leaves scattered on the ground.
Instance Scale + Apply Scale	Scales the real leaf objects of the Instance Collection by this factor and applies the scale. It changes the visible leaves only, so check Physical Bounds afterwards.
Gravity	Pulls flying leaves down, as in Physical Leaves.
Wind Visualizer	Viewport-only arrow above the centre of the ground showing wind direction, strength and lift. It grows with wind speed and is never rendered.

Physical Bounds Properties

Control	What it does
Show Physical Bounds	Draws the contact disk of every leaf in the viewport. Off = nothing is drawn or computed.
Bounds Resolution	Vertices per disk: 3 = cheapest triangle; 64 = smooth circle, slower with many leaves.
Physical Bounds Radius	Disk radius as a multiple of Leaf Size. Raise it if leaves overlap or sink into each other; lower it if they hover apart.
Bounds Thickness Ratio	Disk thickness relative to Leaf Size. Larger = leaves rest further

Control	What it does
	apart along their normal.

Ground Scatter

Control	What it does
Use Ground Collection	On: leaves use the Ground Collection. Off (default after import): they use the single Ground Object.
Ground Object	The ground mesh the leaves are scattered on and land back on. The active mesh goes here on import.
Ground Collection	Several ground objects that work together (Use Ground Collection on). Use it when leaves should fall from one surface onto another, see Ground vs Colliders.
Density	Leaves per square metre of ground.
Seed	Gives a different layout, sizes and rotations.
Leaf Size (Moved in Instances Section in Pre-release version 0.9)	Base size of each leaf; it also sizes the leaf's contact disk.
Random Size	Size variation: each leaf gets Leaf Size × (1 ± Random Size). 0 = all the same size.
Ground Offset	Lifts every leaf slightly off the surface at the start, to avoid clipping.

Masking

Control	What it does
Paint Leaves (Weight Paint)	Opens Weight Paint on the ground with the leaves' vertex group. With several ground objects, it asks which one. With Use Ground Collection off, it paints straight on the Ground Object.
Remove from Vertical Surfaces	On: removes leaves from faces steeper than Vertical Angle. Off: leaves stay off faces steeper than about 40°.
Vertical Angle (degrees)	Steepest slope that keeps leaves: 0 = only flat ground; 90 = vertical. Used while Remove from Vertical Surfaces is on.
Enable Vertex Paint	Places leaves only where the Paint Vertex Group is painted.
Paint Vertex Group	Vertex group on the ground that marks where leaves lie.
Enable Noise Mask	Breaks the leaf cover into natural patches with a noise texture.

Control	What it does
Dissolve Mask	How much ground the patches cover. Higher = more leaves; lower = sparser patches.
Scale	Size of the noise pattern. Higher = smaller, busier patches.
Detail	Noise detail. Higher = more ragged patch edges.
Roughness	Blend between noise layers. Higher = rougher, more broken edges.
Noise Mask Seed	Gives a different patch pattern.

Wind

Control	What it does
Wind (checkbox)	On: wind comes from Angle, Lift and Intensity. Off: the Wind vector is used.
Angle	Wind direction on the ground plane: 0° = +X, 90° = +Y.
Lift	Vertical wind: positive lifts leaves, negative presses them down.
Intensity	Horizontal wind speed. Gusts need it above zero to lift any leaf.
Wind	The wind as an XYZ vector, used when the checkbox is off.

Instances

Control	What it does
Instances (checkbox)	Shows the leaf meshes. Off = the physics still runs, but nothing is drawn.
Instance Object	A single leaf object.
Instance Collection	A collection of leaf variants, picked at random per leaf.
Scale Min / Max	Random size factor per leaf. It scales the visible leaf and its contact disk together (needs Size Scale).
Size Scale	Uses each leaf's size (Leaf Size ± Random Size) for the visible leaf and its disk. Off = every leaf uses size 1.
Offset Rotation	Rotates the visible leaf mesh only, to fix an asset's orientation.

Panels shared with Physical Leaves

Ground Leaves panel	Works like	Differences

Ground Leaves panel	Works like	Differences
Ground Leaf Physics	Physics and flight: Simulation, Aerodynamics Advanced	Gravity sits in General
Flight Turbulence	Physics and flight: Flight Turbulence	None
Landing	Ground and collisions: Leaves Ground Simulation	Leaves are always kept; no Edge Pivot Landing
Collisions	Ground and collisions: Collision, Leaf Contacts	Adds Remove Initial Inside Collider, which deletes leaves that start inside a collider. Colliders Affect Ground Particles is on by default. No Delete on Hit or Fall Off Edges. Avoid Colliders, Avoidance Distance, Avoidance Strength, Ignore Still Colliders and Ground Collider Wake Speed work as in Physical Leaves
Ground Wind | Surface Gusts	Ground Wind	Same controls without the "GW" prefix. Affected Leaves % and Flying Leaves % run from 0 to 1